NKHUB

EDUCATION

DATABASE MANGEMENT SYSTEM (DBMS)

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UNIT – 1 :

Introduction to database management system

Introduction :

As the name suggests, the database management system consists of two parts. They are:

- Database
- Management System

Q.1 Explain Database.

- A database is an organized collection of structured information stored electronically.
- A DBMS(Database Management System) is the software used to manage , store , retrieve, and update this data.
- Product table : contains data on products like product ID , name , price , and stock quantity.

Product ID	Product name	Price	Stock quantity
101	Laptop	$800	50
102	Smartphone	$400	100
103	Headphones	$50	200

Q.2 Explain Data.

- Data is a collection of raw , unorganized facts, numbers, observations , or symbols.
- It is the basic “ingredient” used to build information.
- For example of data simple breakdown
  - ✓ Raw data : 101 , Rahul , 95 (just random values)
  - ✓ Context/structure : Roll Number , Student Name , Marks Scored

- ✓ Information : student Rahul (Roll No.101) scored 95 marks.

Q.3 Explain Record.

- ❖ A record is a horizontal row in a database table that represents a single, complete unit of related data.
- ❖ Each record contains data values for each field defined in the table's schema.
- ❖ Example
  - ✓ In an employees table, a single record contains all the data points for one person – their ID , name , department and salary.
    - Record : [101 , "Rahul Jain" , "Engineering" , $85,000]
    - Fields : Employee_ID , Name , Department , Salary

Employee_ID	Name	Department	Salary
101	Rahul Jain	Engineering	85,000

Q.4 Explain management system.

- ❖ A Database Management System (DBMS) is a software that acts as a bridge between users/applications and a database.
- ❖ It is like a highly organized digital librarian that lets you easily create , store , search , update and protect data doing all the heavy lifting in the background.
- ❖ Popular DBMS include MySQL , Oracle and Microsoft SQL Server.
- ❖ For example
  - ✓ We can use SQL (structured Query Language) which is a common language for interacting with DBMS to perform operations like :
- ❖ Retrieve product information :

- ✓ SELECT ProductName , Price FROM Product Where ProductID = 101;
- ✓ Result : Laptop, $800

Q.5 Explain DBMS (Database Management System).

- ❖ A DBMS (Database Management System) is a software used to create , store and manage data.
- ❖ It acts as a bridge between a database and its users or applications allowing them to easily read , write , update and delete information while keeping it organized and secure.
- ❖ Core functions
  - ✓ **Data storage :** provides a structured way to store everything from simple text to large files.
  - ✓ **Easy retrieval :** Allows users to quickly search and filter large amounts of information using query languages like SQL.
  - ✓ **Data Security :** Prevents unauthorized access and limits what different users can see or change.
  - ✓ **Concurrency Control :** Lets multiple users or apps access and update the same database at the exact same time without causing conflicts.
  - ✓ **Backup and recovery :** Automatically protects against Data loss in the event of system failures or crashes.

Q.6 Basic Definition of DBMS.

- ❖ Data : Data is a collection of raw , unorganized facts, numbers, observations , or symbols.
- ❖ Information : information in a DBMS is organized , processed and meaningful data.
- ❖ Database : A database is an organized collection of structured information stored electronically.
- ❖ DBMS : A DBMS (Database Management System) is a software used to create , store and manage data.

- ❖ Meta Data :
  - ✓ a meta data is simply data about data.
  - ✓ It is holding actual user information (like a customer's name or address).
  - ✓ It contains the rules , descriptions and structure that tell the database how to store and organize that information.

Q.7 explain data vs information.

Aspect	Data	Information
Definition	Data is a collection of raw , unorganized facts, numbers, observations , or symbols.	Processed and organized data.
Example	"42", "john smith", "2023-09-22"	"temperature:42°C", "customer: john smith", "date : 2023-09-22"
Meaning	Lacks inherent meaning	Conveys meaning, context and relevance
Usefulness	Requires processing and interpretation	Valuable for decision making and analysis
Storage	Stored in a DBMS as raw data	May be stored as processed data or derived from data upon a request
Retrieval	Retrieved as is from the database	Retrieved through queries or reports in a structured form
Purpose	Basis for generating information	Supports decision-making and reporting

Q.8 explain Data Redundancy.

- ❖ Data redundancy in a DBMS is the unnecessary repetition of the exact same data in multiple places within a database.
- ❖ For example if a student's address is saved in both a "student" table and a "Library" table, the system wastes storage space and risks update errors.

Q.9 explain data non redundancy.

- ❖ Data non-redundancy means storing each piece of information in exactly one place within a database.
- ❖ Instead of repeating the same data across multiple tables, the database uses simple links or references.
- ❖ This saves storage space and ensures that if you update a fact it automatically updates everywhere.

Q.10 Explain Data Inconsistency.

- ❖ Data inconsistency in a DBMS happens when different copies of the same information in a database contradict each other.
- ❖ It means that the system contains conflicting , out-dated , or mismatched records.
- ❖ For example imagine a bank stores your home address in both a "Customer" table and an "Account" table.
  - ✓ You move and update your address in the customer table, but the system fails to update the Account table.
  - ✓ Now if you check your profile it shows your new address but a bank statement shows your old address.
  - ✓ This mismatched information is data inconsistency.

Q.11 Explain Data integrity.

- ❖ Data integrity in DBMS simply means your data is accurate , complete and trustworthy.
- ❖ It ensures the information is correct and doesn’t get messed up when you save , update , or retrieve it.
- ❖ Think of it as a bouncer that keeps bad or illogical information out of your system.

Q.12 Explain Data Security.

- ❖ Data security in DBMS is the practice of protecting a database from unauthorized access, misuse or damage.
- ❖ It ensures that only the right people can see or change the data keeps the information accurate and prevents hackers from stealing it.

Q.13 Explain Database Authorization.

- Database authorization is the process of giving specific permissions or rights to users, determining exactly what data they can see and what actions they can perform.
- While authentication verifies who you are (like entering password), authorization decides what you are allowed to do after you login.
- Authorized users : have permissions (rights) to access data into database.
- Un-authorized users : do not have permissions (rights) to access data into database.

Q.14 Explain Database Consistency.

- Database Consistency in a DBMS means the system strictly follows predefined rules and laws.
- It ensures any data written to the database is valid and that a transaction cannot secretly create , destroy, or mismatch information, leaving the data accurate both before and after any changes.
- Here are few simple ways to understand it:
  - ✓ **No math Error :** if you transfer $100 from Account A to Account B, consistency ensures the $100 is perfectly subtracted from Account A and added to Account B. The total amount of money in the system can't magically change or vanish.
  - ✓ **Following the rules :** if the database has a rule that an employee's age can't be negative or that an order must have a valid customer ID, consistency prevents bad data from ever being saved.
  - ✓ **No Half-Finished Updates :** If an error happens halfway through saving the database completely reverses any

changes made so the data is not left in a confusing, broken state.

Q.15 Explain Application of DBMS.

Sector	Use of DBMS
Banking	For Customer Information, Account Activities , payments , deposits, loans etc…
Airlines	For reservations and schedule information.
Universities	For student information, course registrations , colleges and grades.
Telecommunication	It helps to keep call records, monthly bills , maintaining balances, etc…
Finance	For storing information about stock, sales and purchases of financial instruments like stocks and bonds.
Sales	Use for storing customer, product & sales information.
Manufacturing	It is used for the management of supply chain and for tracking production of items. Inventories status in warehouses.
HR Management	For Information about employees , salaries , payroll , deduction , generation of paychecks, etc…

Q.16 purpose of database systems.

- The core purpose of a Database Management system (DBMS) is to provide a reliable , structured and secure interface for storing, modifying and retrieving data.
- It bridges the gap between raw data and end-user applications , ensuring that information remains consistent, accessible and protected against unauthorized access or system failures.
- Key functions of DBMS include:
  - ✓ **Data Integrity and Consistency :** Enforces strict rules(like primary and foreign keys) so that data remains accurate and is not duplicated across files.

- ✓ **Concurrent Access :** Allows multiple users or applications to interact with the database at the exact same time without causing conflicts or data corruption.
- ✓ **Security and Access Control:** Protects sensitive information by enforcing role-based permissions, ensuring users can only view or edit what they are authorized to.
- ✓ **Backup and Recovery:** Automatically handles data backups and ensures that incomplete or interrupted operations do not leave the database in a broken state.

Q.17 difference between File system vs DBMS.

File System	DBMS
It is a software system that manages and controls the data files in a computer system.	It is a software system used for creating and managing the databases. DBMS provides a systematic way to access , update and delete data.
File system does not support multi user access.	Database management system supports multi user access.
Data consistency is less in the file system.	Data consistency is more due to the use of normalization.
File system is not secured.	Database management system is highly secured.
In the file system data redundancy is high.	In DBMS data redundancy is low.
No data backup and recovery process is present a file system.	There is a backup recovery for data in DBMS
Handling of a file system is easy.	Handling a DBMS is complex
Cost of a file system is less than the DBMS	Cost of database management system is more than the file system.
If one application fails it does not affect other application in a system.	If the database fails it affects all application which depends on it.
In the file system, data can not be shared because it is distributed in different files.	In DBMS data can be shared as it is stored at one place in a database.

Example: NTFS (New technology file system), EXT(Extended file system), etc.	Example: Oracle, MySQL, MS SQL server , DB2, Microsoft Access, etc…

Q.18 what is schema full explanation.

- ❖ A database schema is the "blueprint" or skeleton of a database.
- ❖ It defines the structure of how data is organized, including table names, columns, data types (like text or numbers), and the rules connecting different tables, without containing any actual data.
- ❖ Think of a schema like the floor plan of an empty house.
- ❖ It dictates exactly where the rooms (tables) are, what goes in them (columns), and how they connect to each other, but it doesn't contain any furniture (the data) yet.
- ❖ There are three levels of schema :

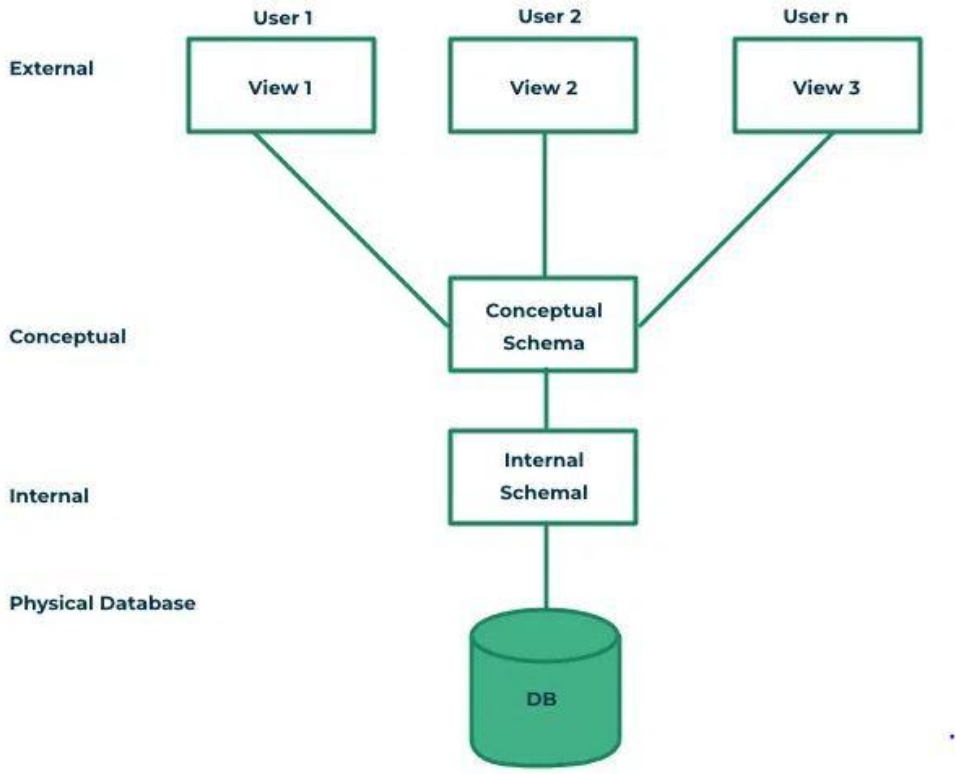


Figure of three level schema

❖ **Physical Schema:** The lowest level. It deals with how data is actually saved on the computer's hard drive (files, indexes, and storage space).

❖ **Logical Schema:** The middle level. It shows the big picture of data—what tables exist, what the columns are, and how tables relate to one another (this is what most developers work with).

❖ **View (External) Schema:** The highest level. It customizes how different users or apps see the database. For example, a cashier sees only item prices and stock, while a store manager sees profit and employee records.

❖ Why is a schema important?

❖ **Data Organization:** It keeps information orderly and prevents duplicates.

❖ **Rules & Integrity:** It enforces rules (e.g., a customer's phone number cannot be text, or a user ID must be unique).

❖ **Security:** It allows different people to see only the parts of the database they are allowed to see.

Q.19 What is Instance full explanation.

❖ An instance in DBMS is the actual data stored in your database at one specific moment.

❖ It is a "snapshot" of your current information.

❖ Because data is constantly added, updated, or deleted, an instance changes frequently, representing the active, working state of the database.

❖ A Real-World Example

- ✓ Imagine a table in a school database designed to hold student information.
- ✓ The Schema:
- ✓ The structure (table and column names) of your database:
  - Table Name: Students
  - Columns: Roll_Number, Name, Grade

❖ The Instance (Morning Snapshot):

- ✓ At 9:00 AM, you open your database and see these two students:
  - 1, Amit, A
  - 2, Kuldip, B

❖ The Instance (Evening Snapshot):

- ✓ At 5:00 PM, a new student joins. You insert their details. The blueprint (schema) is exactly the same, but the instance (the actual data) has changed to this:
  - 1, Amit, A
  - 2, Kuldip, B
  - 3, Naitik, A

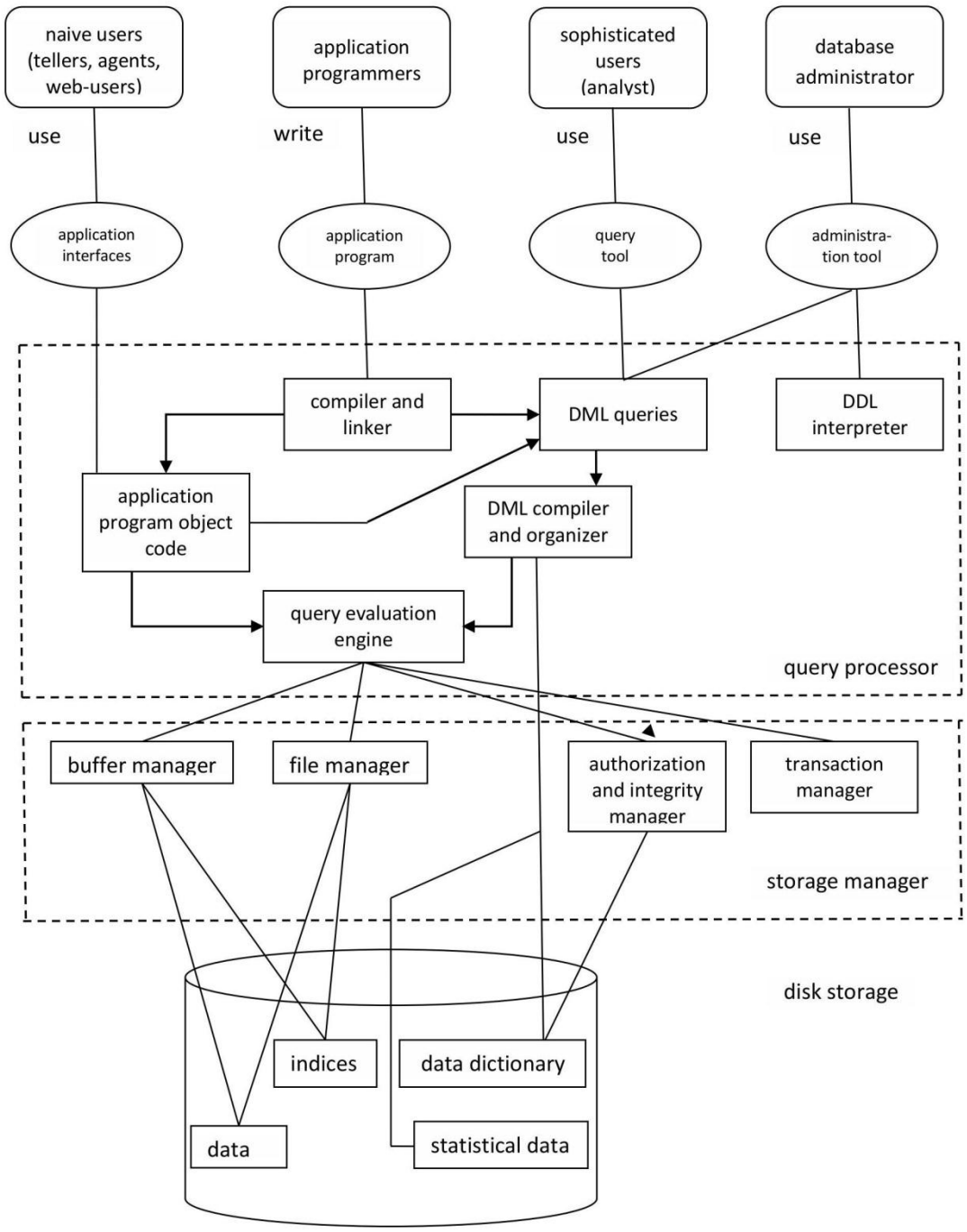
❖ **Key Characteristics of an Instance**

- ✓ **Time-Sensitive:** It represents the database state at exactly the time you look at it.
- ✓ **Dynamic:** It changes whenever records are created, read, updated, or deleted.

Q.20 Explain Difference between schema and instance.

Schema	Instance
It is overall description of the database.	It is the collection of information stored in a database at a particular moment.
Schema is same for whole database.	Data in instances can be changed using addition,deletion,updation.
Does not change frequently	Changes frequently.
Defines the basic structure of the database i.e. how the data will be stored in the database.	It is the set of information stored at a particular time.
Initial state is always empty.	Initial states have always some data.

Q.21 Explain Database System Architecture / Component and function of DBMS.



Database System Architecture

❖ **Database system architecture** is the structural blueprint that defines how a database management system (DBMS) is designed, organized, and connected to its users.

- ❖ Think of it as the construction plan of a library: it dictates where the books (data) are stored, how the librarians (the system) organize them, and how visitors (users) can request and read them.
- ❖ Database architecture is generally explained through two main perspectives: **Tier Architecture** (how the software connects to hardware across a network) and **Three-Level Schema Architecture** (how data is viewed internally versus by the end-user).
- ❖ There are three partitions of the database architecture levels.
  - ✓ **1. Tier Architecture (The Physical Setup)**
  - ✓ **2. Three-Level Schema Architecture (The Logical View)**
  - ✓ **3. Core Internal Components of a DBMS**

1. Tier Architecture (The Physical Setup) :

- ❖ This perspective explains how the user, the application logic, and the physical database are split across different computers or servers.
  - ✓ **1-Tier Architecture**: The user interface, business logic, and database all live on the **same machine**.
    - *Analogy*: Writing notes in a personal notebook. You are the user, and the data is directly in front of you.
    - *Example*: Using Microsoft Access or SQLite locally **on your laptop.**
  - ✓ **2-Tier Architecture:** A basic client-server setup. The user interface sits on your device (client), and it communicates directly with the database sitting on a remote computer (server).
    - Analogy: Calling a restaurant directly to place an order.

- Example: A desktop library app in a school connecting directly to the school's main database server.

✓ **3-Tier Architecture:** The most common structure for modern web applications. An extra layer—the Application Server—is placed between your device and the database to handle business rules and security.

- **Analogy:** Ordering food through a delivery app (the middle layer). You talk to the app, the app talks to the restaurant kitchen (database), and it brings the food back to you.
- **Example:** Logging into Amazon or Netflix. Your browser talks to their application servers, which fetch your data from the database.

2. Three-Level Schema Architecture (The Logical View)

❖ Also known as the ANSI/SPARC architecture, this framework breaks down how data is structured conceptually to keep the physical storage hidden from everyday users.

✓ **External Level (View Level)**:

- This is the **highest level** and is what the user actually sees. Different users see different views of the exact same database depending on their access rights.
- **Example:** In a university database, a student sees their grades, a professor sees the grading portal, and a billing officer sees tuition payments.

✓ **Conceptual Level (Logical Level):**

- This is the middle level that describes what data is actually stored and how the data pieces relate to one another. It defines tables, columns, and data types without worrying about hard drives.

- **Example:** A blueprint stating: "The database contains a Students table with a Student_ID (number) and a Name (text)."

✓ **Internal Level (Physical Level):**

- This is the lowest level that deals with how the data is physically saved on hard drives or SSDs. It handles files, data compression, blocks, and indexing mechanisms.
- Example: The system deciding that a specific table row will be stored as binary data on Sector 4 of an encrypted hard disk drive.

3. Core Internal Components of a DBMS

❖ Behind the scenes, the database system architecture relies on major internal software modules to process instructions:

✓ **Query Processor:** Acts like a translator. It takes your code (like SQL commands), checks it for errors, optimizes it for speed, and converts it into actions the computer can execute.

✓ **Storage Manager:** Acts like a digital filing clerk. It sits between the query processor and the hard drive to handle storing, retrieving, and updating raw data files safely.

✓ **Transaction Manager:** The safety guard. It makes sure the data stays accurate and complete, ensuring operations finish perfectly even if the power suddenly goes out or if thousands of users try to update the exact same file simultaneously.

Q.22 what are the components of DBMS ?

❖ A database management system(DBMS) consists of five fundamental components working together to store, retrieve and manage data efficiently :

✓ Hardware
✓ Software
✓ Data
✓ Procedures
✓ People

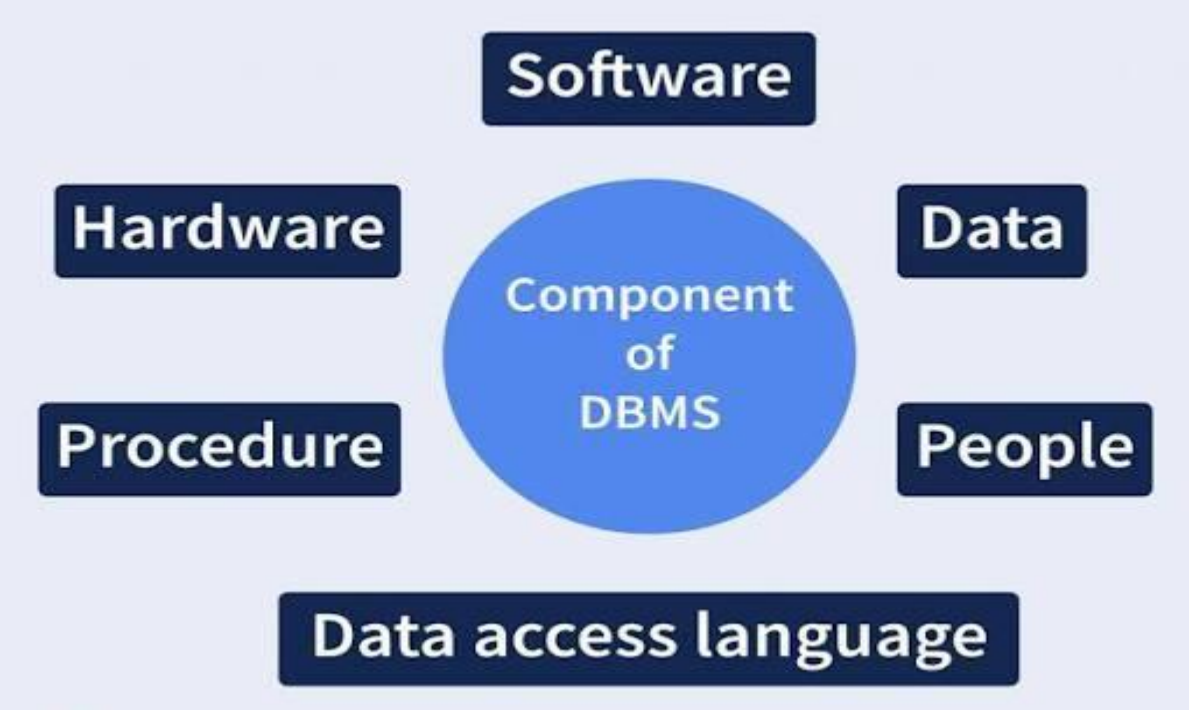


Figure : components of DBMS

❖ Learn how these core parts work together

- ✓ Hardware : the physical computer systems , servers , storage disks and network devices used to house and access the database.
- ✓ Software : the core DBMS programs , operating systems and application program used to manage and process data.
  - This includes the database Engine(executes queries), Query processor (translates commands) and Data dictionary(store metadata).
- ✓ Data : the actual raw information (user data) and "data about data" (meta data) stored within the system.
- ✓ Procedures : the pre-defined rules, instructions and best practices for installing , maintaining , backing up and securing the database.
- ✓ People : the users who interact with the system.
  - This includes database administrators (DBAs), database designers , application developers and end-users.

Q.23 Advantages & Disadvantages of DBMS

❖ **Advantages of DBMS**

✓ **Reducing data redundancy :**

- The file based data management systems contained multiple files that were stored in many different locations in a system or even across multiple systems.
- Because of this, there were sometimes multiple copies of the same file which lead to data redundancy.
- This is prevented in a database as there is a single database and any changes in it is reflected immediately. Because of this, there is no chance of encountering duplicate data.

✓ **Sharing of Data :**

- In a database the users of the database can share the data among themselves.
- There are various levels of authorization to access the data and consequently the data can only be shared based on the correct authorization protocols being followed.
- Many remote users can also access the database simultaneously and share the data between themselves.

✓ **Data integrity :**

- Data integrity means that the data is accurate and consistent in the database.
- Data integrity is very important as there are multiple database in DBMS.
- All of these databases contain data that is visible to multiple users.

- So it is necessary to ensure that the data is correct and consistent in all the databases and for all the users.

✓ **Data security :**

- Data security is very important concept in a database.
- Only authorized users should be allowed to access the database and their identity should be authenticated using a username and password.
- Unauthorized users should not be allowed to access the database under any situation as it violates the integrity constraints.

✓ **Privacy :**

- The privacy rule in a database means only the authorized users can access a database according to its privacy constraints.
- There are levels of database access and a user can only view the data he is allowed to.
- For example – in Social networking sites, access protocols are different for different accounts a user many want to access.

✓ **Backup and Recovery :**

- Database management system automatically takes care of backup and recovery.
- The users do not need to backup data periodically because this is taken care of by the DBMS.
- Moreover, it also restores the database after a crash or system failure to its previous condition.

✓ **Data consistency :**

- Data consistency is ensured in a database because there is no data redundancy.
- Moreover any changes made to the database are immediately reflected to all the users and there is no data inconsistency.

❖ Disadvantages of DBMS

✓ Increased costs :

- One of the disadvantages of DBMS is database systems require sophisticated hardware and software and highly skilled personal.
- The cost of maintaining the hardware, software and personal required to operate and manage a database system can be substantial.
- Training, licensing, and regulation compliance costs are often overlooked when database systems are implemented.

✓ Complexity :

- A DBMS fulfil lots of requirement and it solves many problems related to database.
- But all these functionalities make DBMS extremely complex software.
- Developers , designers, DBA and End users of database must have complete skills of DBMS if they want to use it properly.
- If they do not understand this complex system then it may cause loss of data or database failure.
- As it requires lots of management, so its complexity becomes an issue and very big disadvantage of DBMS.

✓ Technical staff requirement :

- Organizations have many employees working for them and these employees can perform many other tasks too that are not in their domain but it is almost impossible for them to work on DBMS.
- A dedicated team of technical staff is required who can handle DBMS and as a result, Company have to pay handsome salary to them too.

- ✓ **Database Failure :**
  - Data is the key for any organization if data is lost then whole organization will collapse.
  - And as we know that in DBMS all the files are stored in single database so chances of database failure become more.
  - Any accidental failure of component may cause loss of valuable data.
  - This is really a big question mark for big firms and they are continuously working to solve this issue.
- ✓ **Huge size :**
  - DBMS is made to handle extremely huge data and queries, but due its complexity, DBMS has become huge in size.
  - As a result, it requires lots of space and memory to run its application efficiently.
- ✓ **Difficult backup and recovery :**
  - Database is processed and handled by several users concurrently so it becomes very difficult to get the exact state of database at the time of failure.
  - It creates many problems and users get confused what to do next.
  - So it is very difficult to take backup and recover the database as just one application program may pose serious problems.

Q.24 Explain Database Users or its types.

- ❖ Database users are individuals or systems that interact with a database management system(DBMS) to access , update or manage data.

- ❖ They vary by technical expertise and access privileges , ranging from everyday people making transactions to administrators governing the entire system.
- ❖ There are seven types of database users in database management system.
  - ✓ Database administrator (DBA)
  - ✓ Naive / Parametric End Users
  - ✓ Sophisticated Users
  - ✓ Casual / Temporary Users
  - ✓ Application Programmers
  - ✓ Database Designers
  - ✓ Specialized Users
- ❖ Database Administrator (DBA) :
  - ✓ The DBA is the primary custodian and manager of the database system.
  - ✓ They are responsible for authorizing access, monitoring performance, maintaining security, and ensuring data recovery in case of system failures.
- ❖ Naive / Parametric End Users :
  - ✓ These are unsophisticated users with little to no technical knowledge of the underlying DBMS.
  - ✓ They interact with the database daily through pre-built applications or interfaces.
  - ✓ Examples : Railway or airline ticket booking personnel , bank tellers and customers shopping on e-commerce websites.
- ❖ Sophisticated Users :
  - ✓ These are power users(such as engineers, scientists or business analysts) who are highly familiar with database structures.
  - ✓ They interact with the database directly by writing complex SQL queries rather than using pre-programmed interfaces.

- ❖ Casual / Temporary Users :
  - ✓ These users occasionally access the database, but each time they do, they may require different information.
  - ✓ They often use query languages to retrieve data and generally hold management roles within an organization.
  - ✓ Examples : middle or higher-level managers checking monthly sales summaries.
- ❖ Application Programmers :
  - ✓ These are software engineers and developers who write code to create custom applications and interfaces.
  - ✓ They design programs that allow Naive Users to easily query and update the database Without needing to Know SQL.
- ❖ Database Designers :
  - ✓ These users act the architects of the database.
  - ✓ They are responsible for identifying the exact data that needs to be stored and choosing the appropriate tables, constraints, and structural relationship before the database is actively built.
- ❖ Specialized Users :
  - ✓ These are highly specialized programmers who write custom database applications that do not fit into the traditional data –processing framework.
  - ✓ Examples : developers creating artificial intelligence models, expert system, or CAD (Computer –Aided Design) systems.

Q.25 Explain Role of DBA (database Administrator).

- ❖ A database Administrator (DBA) is an IT professional responsible for installing, configuring, securing, and maintaining a DBMS.
- ❖ They ensure databases run efficiently, manage data access, and prevent data loss.

- DBAs balance system performance, data integrity, and availability for an organization's application.
- The core roles and responsibilities of a DBA in a DBMS include:
  - ✓ 1.Database Design and implementation
    - Schema Definition : Collaborating with developers to design and define logical and physical database schemas, tables, and relationship.
    - Storage Planning: Deciding how and where data is stored to optimize system resources and support organizational requirements.
  - ✓ Performance monitoring and Optimization
    - Query Tuning: Identifying performance bottlenecks and optimizing SQL queries and indexes to ensure fast data retrieval.
    - System Monitoring: Continuously tracking metrics like transaction volume, concurrency, and response times.
  - ✓ Data Security and Access Control
    - User Management: Creating and managing user accounts, assigning roles, and revoking permissions to ensure least-privilege access.
    - Compliance: Implementing encryption, auditing, and compliance frameworks to protect sensitive data from unauthorized access or breaches.
  - ✓ Backup and Disaster Recovery
    - Backup Strategies: Establishing periodic data backup schedules and retention policies to safeguard against hardware failure or corruption.
    - Recovery Operations: Developing and regularly testing disaster recovery procedures to ensure seamless data restoration.

- ✓ Maintenance and Upgrades
  - Patch Management: Installing necessary updates, patches, and version upgrades to the DBMS software.
  - Data Migration: Planning and executing the safe transfer of data during system upgrades or changes in the technology stack.

Q.26 Explain Function of DBMS(Database Management System).

- ❖ A Database Management System (DBMS) is software that acts as an interface between databases and end-users or programs, enabling users to create, read, update, and delete data.
- ❖ It maximizes efficiency by organizing data and automating complex tasks to maintain a reliable digital environment.
- ❖ Core Functions of a DBMS includes:
  - ✓ Data Storage &Manipulation: Stores, retrieves, and updates data efficiently using structured formats and query languages like SQL.
  - ✓ Security & Access Control: Protects data from unauthorized access through user authentication and customizable permissions.
  - ✓ Concurrency Control: Manages simultaneous access by multiple users, preventing conflicts and data inconsistencies.
  - ✓ Backup & Recovery: safeguards against data loss by performing routine backups and restoring the database to a consistent state following system failures.
  - ✓ Data Integrity: Enforces rules and constraints (like primary and foreign keys) to ensure data accuracy and validity across the system.